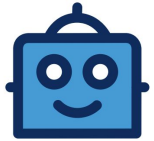
What are the architectural limitations of standard Retrieval-Augmented Generation (RAG) when applied to complex scientific inquiries, and how can multi-agent orchestration mitigate these shortcomings?

query



answer and cite

Standard RAG architectures typically treat complex questions monolithically and perform retrieval in a single step, which frequently yields incomplete or marginally relevant evidence when dealing with multi-faceted scientific topics [1].

To mitigate this, multi-agent orchestrations can be applied to explicitly decompose complex user queries into semantically distinct sub-questions, enabling targeted, parallel retrieval for each aspect of the query [2].

Furthermore, ..

retrieval from corpus

SQuAI: Scientific Question-Answering with Multi-Agent Retrieval-Augmented Generation

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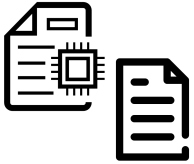
Abstract

We present SQuAI (https://squai.scads.ai/), a scalable and trustworthy multi-agent retrieval-augmented generation (RAG) framework for scientific question answering (QA) with large language models (LLMs). SQuAI addresses key limitations of existing RAG systems in the scholarly domain, where complex, open-domain questions demand accurate answers, explicit claims with citations, and retrieval across millions of scientific documents. Built on over 2.3 million full-text papers from arXiv.org, SQuAI employs four collaborative agents to decompose complex questions into sub-questions, retrieve targeted evidence via hybrid sparse-dense retrieval, and adaptively filter documents to improve contextual relevance. To ensure faithfulness and traceability, SQuAI integrates in-line citations for each generated claim and provides supporting sentences from the source documents. Our system improves faithfulness, answer relevance, and contextual relevance by up to +0.088 (12%) over a strong RAG baseline. We further release a benchmark of 1,000 scientific question–answer–evidence triplets to support reproducibility. With transparent reasoning, verifiable citations, and domain-wide scalability, SQuAI demonstrates how multi-agent RAG enables more trustworthy scientific QA with LLMs.

1 Introduction

Large Language Models (LLMs) have shown strong performance on tasks such as multi-step reasoning [34], summarization [37], and open-domain question answering (QA) [31]. Nonetheless, LLMs often generate hallucinations, i.e., factually incorrect or misleading information [13]. One approach to mitigating hallucinations is retrieval-augmented generation (RAG), which combines traditional information retrieval with LLMs [18]. In RAG, relevant documents are retrieved from external sources and used to ground the model's responses in curated, up-to-date information while reducing the likelihood of unsupported claims.

Scientific QA is a challenging setting, where hallucinations can have especially severe consequences. Unlike general-domain QA, it requires not only factually correct answers but also precise scientific terminology, long-form reasoning, and integration of evidence from multiple sources. Although RAG reduces hallucinations by grounding answers in retrieved documents, standard RAG methods treat questions monolithically and retrieve in a single step, often yielding incomplete or only marginally relevant evidence. Verifiability of LLM-generated answers is also crucial, and as proposed by Gao et al. [11], fine-grained in-line citations help users trace claims back to their sources and assess their reliability.



MA-RAG: Multi-Agent Retrieval-Augmented Generation via Collaborative Chain-of-Thought Reasoning

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To address these challenges, we propose **MA-RAG**, a modular, *training-free* Multi-Agent framework that performs step-by-step reasoning across the entire RAG pipeline (Figure 1(d)). MA-RAG views the RAG process as a collaborative effort among specialized agents, each responsible for a specific subtask such as query disambiguation and task decomposition (Planner), document retrieval (Step Definer and Retrieval Tool), evidence extraction (Extractor), and answer synthesis (QA Agent). Rather than invoking all agents uniformly, MA-RAG adopts an *on-demand* strategy, calling only the necessary agents depending on the ambiguity and complexity at each step. Each agent is guided by chain-of-thought prompting, enabling explicit intermediate reasoning that improves interpretability and task alignment. For instance, agents can decompose an underspecified query into concrete sub-questions, retrieve documents tailored to each subtask, distill targeted evidence from multiple sources, and compose coherent answers by integrating dispersed information.